

THE LEAST COUNTEREXAMPLE TO CONJECTURE 4.3 OF DOLFI, HAFEZIEH, AND SPIGA

ABSTRACT. Dolfi, Hafezieh, and Spiga conjectured that if a positive integer n has $\ell \geq 3$ distinct prime divisors and $\gcd(2^n - 1, n) = 1$, then $2^n - 1$ has at least 2^ℓ distinct prime divisors. We observe that the conjecture is false as stated and prove that its least counterexample is $n = 170$:

$$\gcd(2^{170} - 1, 170) = 1, \quad \omega(2^{170} - 1) = 7 < 8 = 2^{\omega(170)}.$$

The factorization itself appears in prior data; the point of the note is to identify its consequence for the conjecture and prove minimality. Since 170 is even, this does not address the odd-exponent case motivating the original group-theoretic construction.

1. THE COUNTEREXAMPLE

For a positive integer m , let $\omega(m)$ be the number of distinct prime divisors of m , and let $\tau(m)$ be the number of positive divisors of m . Conjecture 4.3 of Dolfi, Hafezieh, and Spiga [2] reads, in full:

“Let n be divisible by ℓ distinct primes and assume $\gcd(2^n - 1, n) = 1$. If $\ell \geq 3$, then $2^n - 1$ is divisible by at least 2^ℓ distinct primes.”

That is,

$$\gcd(2^n - 1, n) = 1, \quad \omega(n) = \ell \geq 3 \quad \implies \quad \omega(2^n - 1) \geq 2^\ell,$$

with no hypothesis of oddness on n . Immediately before the conjecture the same authors write: “Except when $\ell \leq 2$, we have no examples where $2^n - 1$ has exactly $2^{|\pi(n)|} - 1$ distinct prime divisors.” Here $|\pi(n)| = \omega(n)$, and the exponent exhibited below is such an example.

Theorem 1. *The least positive integer violating Conjecture 4.3 of [2] is*

$$n = 170 = 2 \cdot 5 \cdot 17.$$

More precisely,

$$\gcd(2^{170} - 1, 170) = 1 \quad \text{and} \quad \omega(2^{170} - 1) = 7.$$

We use the following specialization of [2, Lemma 4.2].

Lemma 2. *If $\gcd(2^n - 1, n) = 1$, then*

$$\omega(2^n - 1) \geq \tau(n) - 1.$$

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Proof. For every divisor $d > 1$ of n , one has $\Phi_d(2) > 1$; choose a prime $r_d \mid \Phi_d(2)$. Then $r_d \mid 2^n - 1$, so the gcd hypothesis gives $r_d \nmid n$, and hence $r_d \nmid d$. We claim that $\text{ord}_{r_d}(2) = d$. Since $r_d \mid \Phi_d(2) \mid 2^d - 1$, the order $o = \text{ord}_{r_d}(2)$ divides d ; suppose $o < d$. Modulo r_d , the element 2 is a root of $x^o - 1 = \prod_{e \mid o} \Phi_e(x)$ and therefore of some $\Phi_e(x)$ with $e \mid o$, and $e \leq o < d$ forces $e \neq d$. Thus e and d are distinct divisors of d , so $\Phi_e \Phi_d$ divides $x^d - 1$, and 2 is a repeated root of $x^d - 1$ over $\mathbf{F}_{r_d}$. This is impossible because $x^d - 1$ is squarefree over $\mathbf{F}_{r_d}$ when $r_d \nmid d$. Therefore the primes r_d are distinct as d varies, giving $\tau(n) - 1$ distinct prime divisors of $2^n - 1$. $\square$

Proof of Theorem 1. Direct evaluation of the cyclotomic factors for the divisors of 170 gives

$$(1) \quad \begin{aligned} 2^{170} - 1 = & 3 \cdot 11 \cdot 31 \cdot 43691 \cdot 131071 \\ & \cdot 9520972806333758431 \cdot 26831423036065352611. \end{aligned}$$

The first five factors are prime by trial division, and Proposition 5 gives deterministic primality certificates for the last two. Thus $\omega(2^{170} - 1) = 7$.

Moreover, $2^{170} - 1$ is odd. Since $\text{ord}_5(2) = 4$ and $\text{ord}_{17}(2) = 8$, while neither 4 nor 8 divides 170, neither 5 nor 17 divides $2^{170} - 1$. Hence

$$\gcd(2^{170} - 1, 170) = 1,$$

so 170 is a counterexample.

It remains to prove minimality. Suppose that $n < 170$ satisfies the gcd hypothesis, has $\ell = \omega(n) \geq 3$, and violates the conjectured bound. If n is not squarefree, then

$$\tau(n) \geq 3 \cdot 2^{\ell-1},$$

and Lemma 2 yields

$$\omega(2^n - 1) \geq \tau(n) - 1 \geq 3 \cdot 2^{\ell-1} - 1 \geq 2^\ell,$$

a contradiction. Thus n is squarefree. Since $2 \cdot 3 \cdot 5 \cdot 7 = 210$, we must have $\ell = 3$.

The squarefree products of three distinct primes below 170 are

$$30, 42, 66, 70, 78, 102, 105, 110, 114, 130, 138, 154, 165.$$

Every value divisible by 6 fails the gcd hypothesis because $3 \mid n$ and, n being even, $3 \mid 2^n - 1$ as well; hence $3 \mid \gcd(2^n - 1, n)$. The value 105 fails because $\text{ord}_7(2) = 3 \mid 105$, and 110 fails because $\text{ord}_{11}(2) = 10 \mid 110$. Hence any counterexample below 170 must lie in

$$\{70, 130, 154, 165\}.$$

For each remaining value, the proof of Lemma 2 shows that every prime divisor of $\Phi_d(2)$ has order d ; each such n is squarefree with three prime factors, so $\tau(n) = 8$, and hence the seven nontrivial cyclotomic factors contribute

disjoint sets of primes. The following factorizations show that one such set has at least two members:

n	a factored cyclotomic value $\Phi_d(2)$, $d \mid n$
70	$\Phi_{35}(2) = 71 \cdot 122921$,
130	$\Phi_{130}(2) = 131 \cdot 409891 \cdot 7623851$,
154, 165	$\Phi_{11}(2) = 23 \cdot 89$.

All displayed factors are prime by trial division. Thus $\omega(2^n - 1) \geq 8$ in every remaining case, completing the proof. $\square$

Remark 3 (Scope, source of the wording, and prior data). The counterexample concerns the unrestricted wording of Conjecture 4.3 quoted in Section 1. That wording is the one printed in the preprint version of [2], which is the version we were able to consult; the statement number is corroborated for the version of record only indirectly, by [3], who cite “Conjecture 4.3 on page 351”. If the published version carries an additional hypothesis, then what Theorem 1 refutes is the sentence displayed in Section 1.

The construction in Section 3 of [2] assumes more than the conjecture does: it opens “let n be an odd positive integer such that $n \neq p$, $n_3 = 3$ and $\gcd(p^n - 1, n) = 1$ ”. Since 170 is even and $3 \nmid 170$, Theorem 1 says nothing about that regime, and the group-theoretic construction resting on it is untouched.

The factorization itself is not new. It is classical Cunningham-project data: $2^{170} - 1 = (2^{85} - 1)(2^{85} + 1)$, and both factors are recorded in the tables of Brillhart, Lehmer, Selfridge, Tuckerman, and Wagstaff [1], which OEIS A046800 also lists among its references; A046800 records $\omega(2^{170} - 1) = 7$ [4]. Example 3.8 of Laubacher, Lewis, Ravaglia, and Summers [3] records the six factors in (1) other than 3 by taking $p_1 = 5$ and $p_2 = 17$, and cites Conjecture 4.3 on the following page without connecting the two, their construction being confined to odd exponents. No novelty is claimed for the factorization; the point of the note is its consequence for Conjecture 4.3 and the proof of minimality.

2. PRIMALITY CERTIFICATES

We include short Pocklington certificates for the two large factors in (1).

Lemma 4 (Pocklington criterion). *Let $N > 1$ be odd, and let F be a square-free divisor of $N - 1$ whose prime factorization is known. Suppose that $F^2 > N$ and that an integer a satisfies*

$$a^{N-1} \equiv 1 \pmod{N}, \quad \gcd(a^{(N-1)/q} - 1, N) = 1$$

for every prime $q \mid F$. Then N is prime.

Proof. Let r be a prime divisor of N . For every $q \mid F$, the displayed conditions imply that $q \mid \text{ord}_r(a)$, and hence $q \mid r - 1$. Therefore $F \mid r - 1$, so $r > F > \sqrt{N}$. A composite N has a prime divisor at most $\sqrt{N}$, a contradiction. $\square$

Proposition 5. *The integers*

$$P = 9520972806333758431, \quad Q = 26831423036065352611$$

are prime.

Proof. The integers

$$709, 2437, 2857, 6367, 27953$$

are prime by trial division. In each row of the following table, $F \mid N - 1$, $F^2 > N$, and the displayed witness a satisfies the two conditions in Lemma 4 for every prime $q \mid F$. The rows are read from top to bottom within each of the two certificate chains.

N	F	a
2598660833	$709 \cdot 2437$	3
P	$27953 \cdot 2598660833$	15
114281	2857	6
10285291	114281	2
204710635813423	$6367 \cdot 10285291$	3
Q	204710635813423	3

Successive applications of Lemma 4 prove every non-base entry prime, in particular P and Q . $\square$

Exact verification. Every arithmetic assertion above is a finite computation in exact integer arithmetic on the data printed in this paper, and can be redone from the paper alone: multiply the seven factors in (1) and compare the product with $2^{170} - 1$; confirm the five small factors of (1) and the five base primes 709, 2437, 2857, 6367, 27953 by trial division; for each row of the table above check that $F \mid N - 1$, that $F^2 > N$, and that the displayed witness a satisfies the two conditions of Lemma 4 at every prime $q \mid F$; compute $\gcd(2^{170} - 1, 170)$ and the orders $\text{ord}_5(2) = 4$ and $\text{ord}_{17}(2) = 8$; check that the squarefree products of three distinct primes below 170 are exactly the thirteen values displayed in the proof of Theorem 1, that the three stated exclusions remove all but $\{70, 130, 154, 165\}$, and that the cyclotomic factorizations of $\Phi_{35}(2)$, $\Phi_{130}(2)$ and $\Phi_{11}(2)$ displayed there are correct.

The computation was also re-run from scratch and independently, on a separate machine, by a program that takes the numbers claimed here as its input and recomputes everything else: all 36 of its checks agreed with the statements above. That re-run went beyond the list just given in three ways: it proved P and Q prime a second time by Lucas certificates built independently of the Pocklington chains, it factored $2^n - 1$ completely for each of the four exponents below 170 that satisfy both hypotheses, and it confirmed by mutation that the same predicates report failure when the exhibited factorization or either large factor is perturbed. The program and its transcript are retained in an auxiliary archive, available on request.

Two things the re-run did not settle. A supplementary sweep of the bound of Lemma 2 was carried out only for $n \leq 100$, because factoring $2^n - 1$ for prime exponents near 170 was outside its budget; that sweep is an extra, and the bound is proved above rather than computed. And the bibliographic attributions in Remark 3 are not checkable by machine; they were confirmed by reading the sources. An independent re-run that agrees is evidence that the computation is right; the proofs above, not the program, are what is offered as the argument.

REFERENCES

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- [4] OEIS Foundation Inc., *The On-Line Encyclopedia of Integer Sequences*, Sequence A046800, “Number of distinct prime factors of $2^n - 1$,” accessed August 22, 2026. oeis.org/A046800. The terms displayed on the sequence’s page stop at $n = 101$; the value at $n = 170$ is in the linked b-file, which tabulates $a(n)$ for $n = 0, \dots, 1206$.